

Coding & Solution

Fit Buddy AI – AI Fitness Plan Generator using Gemini Models

Generative AI Project | FastAPI + Google Gemini

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Implementation

FitBuddy AI is implemented as a Python web application with a FastAPI backend, Jinja2 / HTML / CSS frontend, SQLAlchemy with SQLite for storage and a Gemini-powered service layer.

Major modules

- Workout Plan Generation
- Nutrition / Recovery Tip
- Feedback-Based Plan Revision
- Admin Users View

AI integration

User details are converted into prompts. Gemini returns a JSON plan that is validated against a Pydantic schema requiring exactly 7 days. If Gemini is unavailable, a local fallback generator creates the plan so the application keeps working.

Security note

The API key is configured through environment variables (.env) and must not be committed to GitHub. Use .env.example as the template.